From corpora to grammars

Modeling phonological gradience from written data

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1 Overview

Goal

Today we connect two things that are often taught separately:

- **Corpus-based description**: what written data can reveal about phonological patterns
- **Probabilistic grammar**: how those patterns can be formalized as weighted constraints
- **Workflow**: from observable distributions to explicit predictions using an R package

> **Question:** How do we move from written forms in a corpus to a probabilistic phonological grammar?

Plan

This is a mixed seminar: roughly half slides, half live R work.

1. From categorical to probabilistic patterns in phonology
2. What written corpora can and cannot tell us: a case study in Talian
3. How `Fonology` turns written forms into phonological data (Garcia, 2026)
4. How MaxEnt grammar models variable outcomes
5. Hands-on: English and Portuguese; corpus interpretation, tableaux, and model predictions

2 Introduction

Timing

Slides

- Corpus evidence and gradience
- English & Portuguese as case studies
- MaxEnt grammar
- Tableau exercises

R

- Generate phonological datasets
- Extract patterns from written forms
- Visualize distributions
- Run and interpret MaxEnt

Gradience

Phonological descriptions often start with categorical statements:

- ‘This segment is deleted’
- ‘This cluster is repaired’
- ‘This stress pattern is preferre’
- ‘This syllable shape is avoided’

☞ But real linguistic behavior often looks more like:

> Option a is **more likely** than b, and that likelihood depends on X variables

Why corpora?

Corpus → makes gradience 'visible' given large quantity of data

- We can count types or tokens
- We can compare contexts
- We can ask which structures are frequent, rare, or absent
- We can separate a pattern from a few hand-picked examples

Caution: Written data are *not* speech data. The point is not to pretend spelling is phonetics, but to exploit the phonological information encoded in graphemes (e.g., **lexicon** vs **grammar**)

From corpus to grammar

written forms → phonological variables → distributions → formalization

Figure 1: A practical workflow for today

How to implement this? We need the proper tools to work with written data. We'll use an R package for that: `Fonology` (Garcia, 2026). This will help us with grapheme-to-phoneme conversion, phonological variables, and formalized grammars (MaxEnt learning today).

Written evidence

1. What phonological patterns could benefit from **written** corpora?
3. What would written data fail to tell us?

Written evidence: benefits and limitations

Useful

- Segmental sequences (phonotactics)
- Orthographically encoded contrasts
- Syllable-shape proxies
- Stress when recoverable
- Type frequency

Limited

- Phonetic detail
- Gradient acoustic realization
- Sociophonetic variation
- Connected-speech reductions
- Orthographic inconsistencies

☞ No standardized spelling: graphemes *could* reveal some phonological variation (Garcia & Guzzo, 2023)

3 From categories to probabilities

Some key moments in the history of phonology

1948
Probabilistic thinking (Shannon, 1948)

1968
SPE (Chomsky & Halle, 1968)

1976
Autosegmental phonology (Goldsmith, 1976)

1982
Lexical Phonology & Morphology (Kiparsky, 1982)

1993
Optimality Theory (Prince & Smolensky, 2004)

1998
Stochastic OT & GLA (Boersma, 1997)

2001
Usage-based phonology (Bybee, 2001)

2001
Exemplar dynamics (Pierrehumbert, 2001)

2003
MaxEnt (Goldwater & Johnson, 2003)

2022
'70 years of probabilistic phonology' (Pierrehumbert, 2022)

False dichotomies

Theory and data are **not** competing dimensions of analysis

- Phonology has **always** depended on empirical evidence (data points); e.g., Pāṇini (Kiparsky, 2022)
- Traditionally, however, it has *tended* towards **categoricity**
- But variation is part of linguistic knowledge (e.g., sociophonetics, sociolinguistics)
- Quantitative data can make generalizations **testable**

☞ **A model is useful only if we know what it predicts**

> 'Either theory or statistics?' We need **both**: an empirical pattern and an explicit analysis

The three ingredients

For this seminar, the workflow has three ingredients:

1. A phonological question
2. A dataset that makes the pattern measurable
3. A probabilistic grammar that models and predicts the distribution

Today's case study: Talian written corpus → metaphony → MaxEnt

☞ Then we'll apply the same logic to English and Portuguese to compare their **phonotactics**

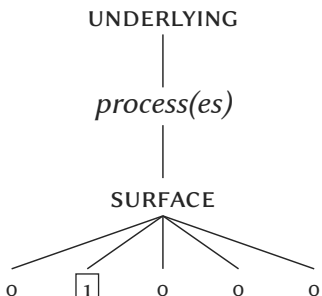
Why a probabilistic grammar?

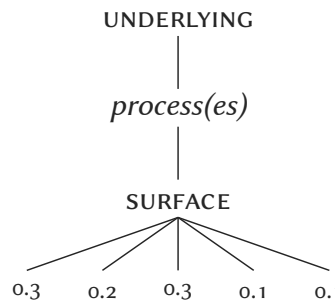
If a pattern is gradient, a categorical grammar is unrealistic:

- It can say which surface form ‘wins’ or surfaces — *brick* vs ~~*bnick*~~
- But it *cannot* say that one form is likely but not guaranteed — ~~*bnick*~~ vs *?blick*
- It cannot directly compare observed frequencies with predicted probabilities

> ☞ We will not ask only ‘which form wins’? We will ask ‘how probable is each form’?

Why a probabilistic grammar? (ii)





4 Constraint-based grammar

From rules to constraints

- The transition from SPE[1] (rules) to OT[2] (constraints) was not abrupt (Paradis, 1987)
- Between the 1960s and the 1990s, several phonological models were proposed
- After SPE, OT is probably the most influential model (Prince & Smolensky, 2004)

Core idea: Universal constraints can be violated (cf. rules in SPE)

[1] The Sound Pattern of English (Chomsky & Halle, 1968).
[2] Optimality Theory (Prince & Smolensky, 1993).

SPE vs (standard) OT: two architectures

Underlying representation

↓

Rule 1: A → B / ____ X

Rule 2: B → C / ____ Y

Rule 3: C → D / ____ Z

↓

Surface representation

Figure 2: **Serial** evaluation in SPE

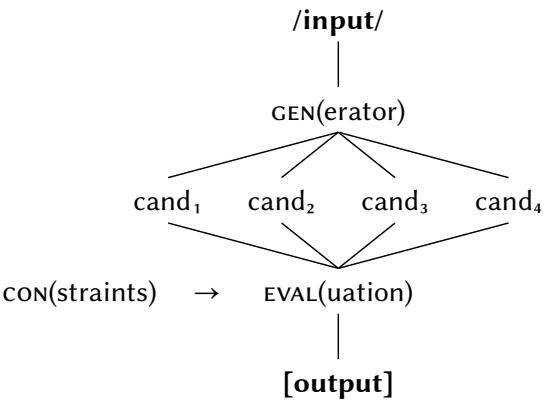


Figure 3: **Parallel** evaluation in OT

SPE vs (standard) OT: two architectures

/pəlis/

↓

Rule 1: ə → Ø

Rule 2: l → l̥ / p ____

↓

[pl̥is]

Figure 4: **Serial** evaluation in SPE

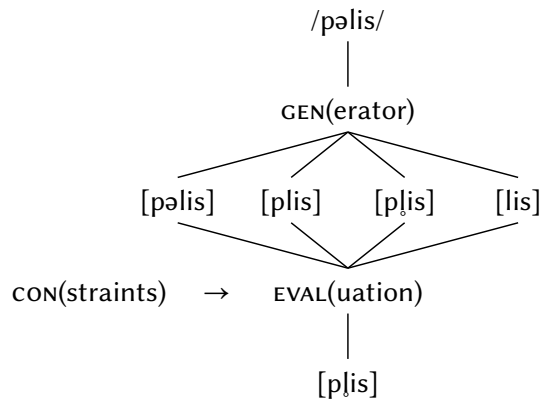


Figure 5: **Parallel** evaluation in OT

Core ideas of OT

1. **Violability**: constraints can be violated, but as little as possible
2. **Ranking**: languages differ in how they rank the same constraints
3. **Parallelism**: the winner is evaluated against the full ranking and the full candidate set

☞ Point (3) above soon proved to be a fundamental problem with OT

> **Why OT?** We're merely using OT here because constraints offer an easier transition into probabilistic approaches such as Maximum Entropy Grammars, which we will discuss in more detail soon

Example: final devoicing

German-like ranking

	bɑd	*VoicedCoda	Ident-IO[vce]
	bɑd	*!	
☞	bɑt		*

English-like ranking

	bae d	Ident-IO(vce)	*VoicedCoda
☞	bæd		*
	bæt	*!	

- **German**: final devoicing: *Bad* → [bɑt] ‘bath’
- Compare with **English**: *bad* → [bæd], *[bæt]

Markedness vs. faithfulness

- OT is built around a tension between two major constraint families
- **Markedness**: prefer well-formed output structures (‘reduce complexity in the output’)
- **Faithfulness**: preserve the relation between input and output (‘don’t change the input’)

Typical faithfulness and markedness constraints:

Faithfulness

- MAX: do not delete
- DEP: do not insert
- IDENT: preserve feature values

Markedness

- ONSET: have an onset
- *CODA: don’t have a coda
- *COMPLEX: don’t have complex sequences (CC)

A concrete conflict

Some languages avoid complex onsets or codas.

- This motivates markedness constraints such as *ComplexOnset or *ComplexCoda
- Different repair strategies change the output relative to the input
- Deletion helps markedness but violates Max
- Epenthesis helps markedness but violates Dep

Example: *Stella* in English: [tɛlə] or [ˈsɛtɛlə] or [esˈtɛla] depending on L1, proficiency level, etc.

Constraints should not be arbitrary

Two classic constraints previously mentioned:

- *Coda: assign one violation for each coda in the output
- Onset: assign one violation for each syllable without an onset

- Languages do not ban onsets across the board
- But codas are frequently **more marked** than onsets
- CV syllables are often treated as less marked than CVC syllables

☞ These constraints are therefore phonetically and typologically motivated

An intuitive model

OT is often intuitive because it resembles a decision problem:

☞ the ‘best’ candidate depends on multiple variables, and no candidate is necessarily ‘perfect’

Constraints as variables:

Constraints can also be thought of as **variables** in a statistical model, as we will see shortly

5 MaxEnt Grammar

MaxEnt

In a Maximum Entropy grammar, every candidate gets a probability.

- Constraints have positive weights
- Violations determine the harmony score h_i
- h_i is converted into an unnormalized score
- Scores are normalized so that probabilities sum to 1

Key difference from classic OT: We are modeling a **probability distribution** over candidates, not just choosing a single winner.

A first MaxEnt tableau

	$w = 1.9$	$w = 0.8$	$w = 0.1$			
/input/	C3	C2	C1	h_i	e^{-h_i}	P_i
a. [a]	0	0	1	0.1	0.905	0.602
b. [b]	0	1	0	0.8	0.449	0.299
c. [c]	1	0	0	1.9	0.15	0.099

Figure 6: A simple MaxEnt tableau

(Goldwater & Johnson, 2003; Hayes & Wilson, 2008; Wilson, 2006)

Interpreting the tableau

The candidate probabilities follow directly from the weights.

- Candidate [a] violates the weakest constraint
- Candidate [b] violates an intermediate constraint
- Candidate [c] violates the strongest constraint

> The order of the columns is visually helpful, but it's the **weights** that do the work

MaxEnt calculation

The calculation has four steps:

1. Multiply each violation by the relevant constraint weight
2. Sum the products for each candidate: h_i
3. Convert the harmony score: e^{-h_i}
4. Divide each value by the sum of all values

$$P(y|x) = \frac{e^{-\sum_{i=1}^{n} w_i C_{i(y,x)}}}{Z(x)} \tag{1}$$

Weighted constraints and gang-up effects

- A candidate may violate a more costly constraint and still be more probable (P_i)
- Notice how candidade [d] violates a less important constraint (C4) — but 5 times, which adds up

	$w = 0.5$	$w = 0.4$	$w = 0.3$	$w = 0.2$	$w = 0.1$				
/input/	C1	C2	C3	C4	C5	h_i	e^{-h_i}	P_i	
[a]	1	0	0	0	1	0.6	0.549	0.331	
[b]	0	1	2	0	0	1	0.368	0.222	
[c]	0	1	1	1	0	0.9	0.407	0.245	
[d]	0	0	0	5	1	1.1	0.333	0.201	

Figure 7: Example of weighted constraints and gang-up effect

MaxEnt: an illustrative example from English and Japanese

- When acquiring English, its phonotatics can often be a challenge. CCCV sequences are an example

	$w = 5$	$w = 3$	$w = 2$	$w = 0.1$			
/CCCVVC/	*Complex	*Coda	Max	Dep	h_i	e^{-h_i}	P_i
[CCCVVC]	1	1	0	0	8	0	0
[CCVC]	1	1	2	0	12	0	0
[CVC]	0	1	3	0	9	0	0
[CV]	0	0	4	1	8.1	0	0
[CVCVCVVCV]	0	0	0	3	0.3	0.741	0.999

Figure 8: Suppose an input like *strike* in English adapted to Japanese *sutoraiko*

Summary

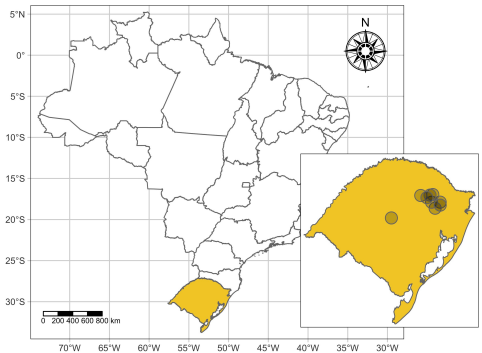
- Probabilistic approaches have existed for a long timcorpuse
- Now they're much closer to the norm: realistic data are rarely categorical
- Constraint-based frameworks are easy to adapt (Noisy Harmonic Grammars, MaxEnt)

☞ These approaches are essentially statisticaly models with phonological assumptions

Next: How to go from **written data** to corpus to phonological patterns to grammar model (MaxEnt)

6 Case study: Talian

Why Talian?



- Brazilian Veneto: Veneto-based koiné in Brazil
- It is also known as **Talian** (Frosi & Mioranza, 1983; 2009)
- Close to Central Veneto (Belloni, 2009)

No standardized orthography

- Graphemes can thus reflect phonological variation more directly than in heavily standardized languages (Garcia & Guzzo, 2023)

Here, we go from **written evidence** to **structured variation** and then to a **probabilistic grammar**

The pattern

- In Veneto varieties, /é ó/ variably raise to [i u] when followed by final /i/ (Walker, 2005; 2010; Zamboni, 1974)
- The trigger is usually morphological: plural /-i/ or second person singular -ɪ
- In Brazilian Veneto, the process is variable, not categorical

Penultimate stress

[ˈovi] ~ [ˈuvi] ‘egg.PL’
[ˈbevi] ~ [ˈbivi] ‘drink.2SG’

Other stress positions

[ˈzoveni] ~ [ˈzuvini] ‘young.PL’
[faˈzoi] ~ [faˈzui] ‘bean.PL’

Why written data work here

Garcia & Guzzo (2023) examine metaphony through the Talian Corpus (Garcia & Guzzo, 2021)

- There is little digitized Talian text, so the corpus required OCR and manual cleanup (Smith, 2007)
- Source materials include book excerpts and newspaper articles
- Once digitized, the corpus can be treated as tidy data for phonological analysis (Wickham, 2014)
- Grapheme-to-phoneme mapping is stable enough to recover the relevant metaphony contexts

Key methodological point: The corpus is *not* a proxy for phonetics. It is a proxy for distributional phonology when the orthography preserves the contrasts we care about.

From corpus to dataset

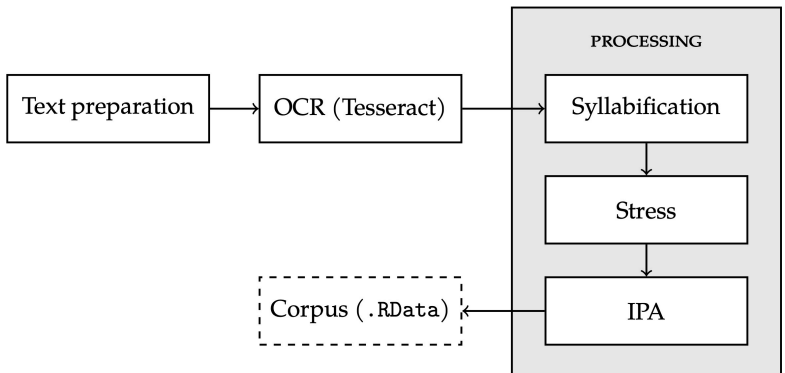


Figure 9: Corpus workflow from source to analyzable data

From corpus to dataset

- Extract all words that could undergo metaphony
- Restrict to polysyllables ending in unstressed /i/ with stressed underlying /e, o/
- Manually verify stress assignment and target-vowel quality
- Focus the quantitative analysis on penultimate-stress words with 2 and 3 syllables

☞ Final analysis sample: 2,095 tokens (490 unique items)

Main empirical asymmetry

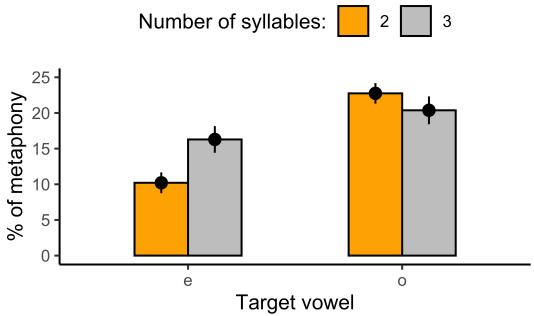


Figure 10: Metaphony by target V and word length

- /o/ undergoes metaphony more often than /e/
- The difference interacts with word length
- 3-σ words favor metaphony more strongly for /e/
- 2-σ words show the clearest advantage for /o/

This already shows that a purely categorical description is too simplistic

Lexicon and grammar

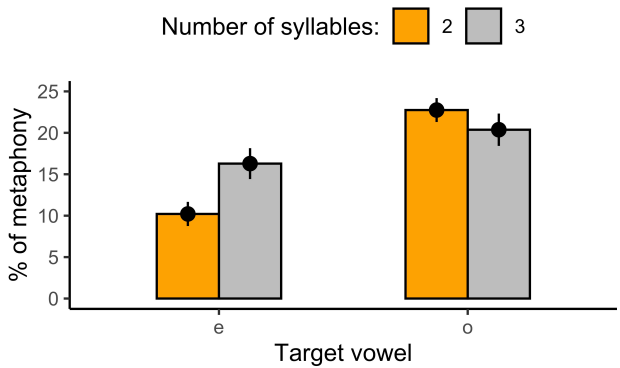


Figure 11: Observed metaphony patterns

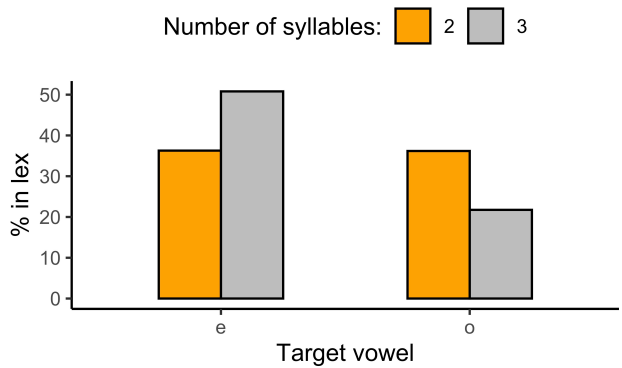


Figure 12: Distribution in the wider corpus lexicon

- The metaphony asymmetries mirror lexical distributions in the corpus
- ☞ A realistic grammar must connect structural constraints with observed probabilities

The puzzle

- The previous results concern general rates of metaphony
- Now let's focus on a more specific question: why does [oi] → [ui] but /ei/ does not surface as [ii]?

Attested metaphony

[faˈzol] 'bean.sg' → [faˈzoi] ~ [faˈzui]
[ˈkori] ~ [ˈkuri]

Blocked outcome

[kaˈvel] 'hair.sg' → [kaˈvei], not [kaˈvii]
[eˈbrei], not [eˈbrii]

Why this is interesting

- [ii] sequences **do** exist in Brazilian Veneto
- But they are tolerated only when inherited, not when newly created by metaphony

Inherited [ii]	**Metaphonic [ii]**
[ˈzii] ‘uncle.PL’	[kaˈvii] impossible
[imbamˈbii] ‘stupid.PL’	[eˈbrii] impossible

Question: How can we formalize this distinction between inherited and newly generated forms?

Comparative markedness

- A single OCP-style ban on identical adjacent vowels is not enough. We need to distinguish:

- $_{N}$OCP: penalizes new identical-vowel sequences
- $_{O}$OCP: penalizes inherited identical-vowel sequences
- $_{O}$LIC: favors spreading/post-tonic licensing of [+hi] (Walker, 2005)
- IDENT[hi]: resists changing vowel height

☞ This is what **Comparative markedness** does (McCarthy, 2003)

> Brazilian Veneto ranks the grammar so that $_{N}$OCP matters more than $_{O}$OCP in the metaphony context

Tableau: metaphony can win

	fa'zo-i	NOCP	OLIC[+hi]	IDENT[hi]	OOCP
	fa'zoi		*!		
☞	fa'zui			*	

Tableau 1: Evaluation of /fa'zo-i/ → [fa'zui]

- Nothing new and identical is created, so NOCP is irrelevant
- The pressure from OLIC can therefore favor metaphony

Tableau: metaphony can be blocked

ka've-i	NOCP	OLIC[+hi]	IDENT[hi]	OOCP
ka'vii	*!		*	
☞ ka'vei		*		

Tableau 2: Evaluation of /ka've-i/ → [ka'vei], not [ka'vii]

- The metaphonic candidate creates a new [ii] sequence
- NOCP blocks that output even though metaphony is otherwise available

From categorical to probabilistic

- The comparative-markedness analysis explains the structural asymmetry
- But the corpus shows **variation**, so we still need a **probabilistic** grammar
- **MaxEnt** will be helpful here (Goldwater & Johnson, 2003; Hayes & Wilson, 2008; Wilson, 2006)

But we don't have experimental data yet... We don't know how frequent metaphony is in spoken Talian yet. But this is not a problem: we can easily simulate different scenarios

From categorical to probabilistic

ka've-i	NOCP ($w = 7.42$)	OLIC[+hi] ($w = 2.82$)	IDENT[hi] ($w = 2.82$)	OOCP ($w = 1.03$)	h_i	e^{-h_i}	P_i
ka'vii	1	0	1	0	10.24	0	0.001
ka'vei	0	1	0	0	2.82	0.06	0.999

fa'zo-i	NOCP ($w = 7.42$)	OLIC[+hi] ($w = 2.82$)	IDENT[hi] ($w = 2.82$)	OOCP ($w = 1.03$)	h_i	e^{-h_i}	P_i
fa'zui	0	0	1	0	2.82	0.06	0.5
fa'zoi	0	1	0	0	2.82	0.06	0.5

Figure 13: Example of MaxEnt tableaux for our data assuming 50% rate of metaphony for /oi/

From categorical to probabilistic

- What if the rate of metaphony for /oi/ isn't 50%?

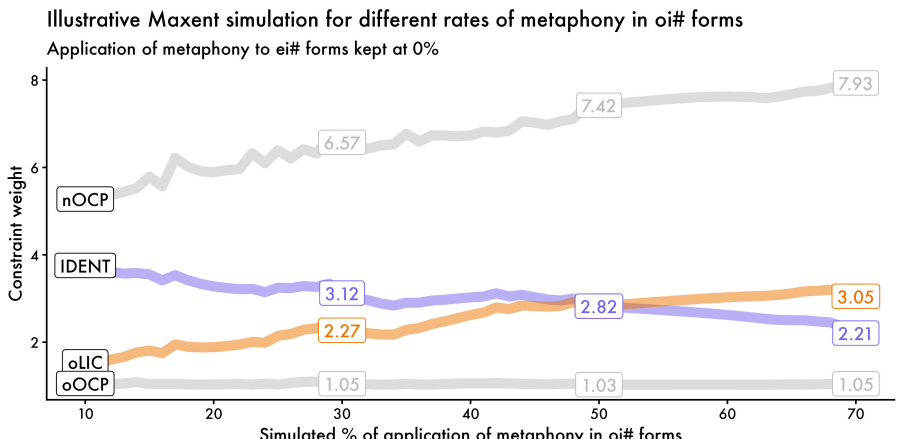


Figure 14: Simulation of how constraint weights change with metaphony rates

Final remarks

- Written corpora can reveal gradient phonological patterns when the relevant structure is recoverable
- `Fonology` gives us a practical bridge from written forms to analyzable phonological variables
- MaxEnt grammar turns corpus distributions into explicit candidate probabilities
- Weighted constraints let us model cumulative effects
- The central workflow is empirical and formal at the same time

Full workflow: written forms → corpus asymmetry → constraints → predicted probabilities

7 R: hands-on

Before we get started

- R is a language designed for data analysis (R Core Team, 2026)
- It's become the standard tool in many areas, including linguistics (also Python, Julia)
- It offers a rich ecosystem, with 20k+ packages/libraries

Our work environment: We can use R with a text editor, an IDE or **online**. Today, we will rely on **posit.cloud**, an online interface that resembles RStudio, the most popular IDE to use R. Positron is a more modern option.

Fonology

- An R package for phonology: we will explore it today

```r
ipa("hospital", lg = "en")
ipa("mañana", lg = "sp")
ipa("enfants", lg = "fr")
ipa("giorno", lg = "it")
ipa("almoço", lg = "pt")
```

ˈhɑs.pɪ.təl
ma.ˈɲa.na
ɑ̃.fɑ̃
ˈdʒor.no
al.ˈmo.so

`ipa()` → Grapheme-to-phoneme conversion is the **most important** function in the package

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